

The video element allows multiple source elements. Source elements can link to different video files. The browser will use the first recognised format, as follows:

```
<video width="320" height="240" controls="controls">
  <source src="video.ogg" type="video/ogg" />
  <source src=" video.mp4" type="video/mp4" />
  <source src=" video.webm" type="video/webm" />
This browser does not support the video element.
</video>
```

Audio

With audio, the case is similar to video. Until HTML5 was launched, there wasn't any uniform standard for playing audio on Web pages. Most of the audio was also played through different plugins like Flash. But HTML5 specifies a standard way to play audio on a Web page by using an audio element. The audio element is used to play sound files and to stream audio.

Currently, HTML5 supports three audio formats for the audio element, as shown in Table 3.

Table 3

Format	IE 8	Firefox 3.5	Opera 10.5	Chrome 3.0	Safari 3.0
Ogg Vorbis	No	Yes	Yes	Yes	No
MP3	No	No	No	Yes	Yes
Wav	No	Yes	Yes	No	Yes

The use of the audio element is demonstrated below:

```
<!DOCTYPE HTML>
<html>
<body>

<audio src=" song.ogg" controls="controls">

This browser does not support the audio element.

</video>

</body>
</html>
```

This example uses an Ogg file and will work in Firefox, Opera and Chrome. To make the audio work in Safari and future versions of Chrome, we must add an MP3 and Wav file.

The audio element allows multiple source elements, which can link to different audio files. The browser will use the first recognised format, as follows:

```
<audio controls="controls">
  <source src="song.ogg" type="audio/ogg" />
```

```
<source src="song.mp3" type="audio/mpeg" />
```

This browser does not support the audio element.

```
</audio>
```

Canvas

To create graphics on a Web page, HTML5 uses the Canvas API. We can draw anything with it and it uses JavaScript. This can improve the performance of our website by avoiding the need to download images off the network. With Canvas, we can draw shapes and lines, arcs and text, gradients and patterns. In addition, Canvas gives us the power to manipulate pixels in the images and even in video. You can add the Canvas element to an HTML page as shown below:

```
<canvas id="myCanvas" width="200" height="100"></canvas>
```

The Canvas element doesn't have capabilities to draw elements. We can achieve this by using JavaScript. All drawings should be inside JavaScript.

```
<script type="text/javascript">
var c=document.getElementById("myCanvas");
var ctx=c.getContext("2d");
ctx.fillStyle="blue";
ctx.strokeStyle = "red";
ctx.fillRect(10,10,100,100);
ctx.strokeRect(10,10,100,100);
</script>
```

The output of the above script is seen in Figure 6.



Figure 6: Output of Canvas

You can draw many objects like arcs, circles, linear/vertical gradients, etc.

HTML5 tools

To operate effectively, all skilled or amateur Web developers/